

# Daniel Alejandro Silva Rojas

## SENIOR SYSTEMS & PERFORMANCE ENGINEER · BACKEND & AI

Caracas, Venezuela · Remote | +58 414-231-7561 | dsrglrm@gmail.com | my-resume-landing.vercel.app | github.com/Dan178A | linkedin.com/in/daniel-alejandro-silva-rojas

### PROFESSIONAL SUMMARY

Systems & Performance Engineer with 5+ years of experience leading low-latency, real-time architectures in C++, Rust, and Python. At Ea2technology (Canada) I design industrial Digital Twins for refinery monitoring and full-duplex WebSocket architectures that reduced latency from 200-500ms to 1-10ms. I complement this with computer vision, scientific computing (NumPy/SciPy), and RAG systems, plus full-stack development (Vue3/Nuxt3, FastAPI, Supabase). Academic training (B.Sc. in Computer Science, LUZ-IUTA) completed in parallel with active professional experience since 2021. Founder of DropAudio CCS, an e-commerce platform in production — a part-time personal project alongside my full-time primary role and selective freelance consulting.

### TECHNICAL SKILLS

**Languages:** Rust, C++, Python, TypeScript, JavaScript (ES6+), SQL

**Systems & Real-Time:** Tokio, Axum, Rocket, full-duplex WebSockets, gRPC, TCP Sockets, Digital Twin architectures

**AI & Scientific Computing:** FastAPI, RAG systems, NumPy, SciPy, OpenCV, PyO3, FlowNet / deep learning

**Data & Cloud:** PostgreSQL, MySQL, Supabase, Redis, MongoDB, AWS (EC2, S3, RDS, Lambda), Docker

**Specialties:** Low-latency architecture, concurrency, memory safety, industrial telemetry, computer vision

### PROFESSIONAL EXPERIENCE

#### Lead Performance & Systems Engineer

*Jul 2021 – Present (full-time)*

Ea2technology — Canada (remote)

- Designed and deployed an industrial Digital Twin system (telemetry) for real-time monitoring of coke drum deformation in a refinery, processing high-frequency sensor data to predict and prevent critical structural failures.
- Designed a low-latency, full-duplex WebSocket architecture for dynamic dashboards, replacing HTTP polling: reduced latency from 200-500ms to 1-10ms and cut bandwidth usage by up to 95%.
- Led the migration of legacy monolithic systems to cloud-native microservices on AWS, optimizing compute-intensive data pipelines with C++ and Rust to ensure memory safety and maximum concurrency under load.
- Led end-to-end technical UI/UX design in Figma, connecting low-level performance with intuitive operator interfaces.

#### Founder & Software Architect

*Jun 2021 – Present (part-time personal project)*

DropAudio CCS — dropaudioccs.com (e-commerce platform in production)

- Designed and launched a complete e-commerce platform (Nuxt 3 SSR + Supabase/PostgreSQL with RLS) to production, hosted on Vercel, consolidating 94+ verified customer reviews.
- Developed a 3-step interactive recommender and technical comparison tool that reduced purchase friction and increased conversion.
- Implemented a custom multi-currency checkout (USDT, Zinli, Pago Móvil) with live BCV/USDT exchange rates and automated Web Push notifications via pg\_cron.
- Built a resilient architecture with a backup catalog for outages and technical SEO/GEO (JSON-LD, sitemap, PWA) to ensure operational continuity.

#### Software Performance Consultant & Algorithm Engineer

*Jan 2023 – Present (occasional/selective consulting)*

Freelance / Independent Consulting · Remote — international client Vessel Inspection and Assessment Corp (VIAAC), maritime/oil sector — role: Senior Developer

- Developed high-performance video stabilization engines using optical flow algorithms and matrix interpolation (Python, NumPy, SciPy, OpenCV) for trajectory smoothing via Euclidean transformations, achieving a 65-80% reduction in jitter depending on the video, processing at 60fps across HD to Full HD resolutions.
- Implemented global motion estimation with FlowNet (distilled deep architectures) to correct unstable video, combining traditional mathematics with modern machine learning; this research-oriented work later served as the foundation for a camera tracking system.

- Designed and deployed TCP Socket and WebSocket communication systems, including custom file-transfer protocols, for high-frequency environments.
- Conducted 4 technology-stack audits for client VIAAC, identifying 3 memory leaks and concurrency bottlenecks in the processing of point clouds of up to 1.2M points (3D meshes of drums); the resulting optimizations reduced a heavy sequential process from 12-14 hours to 5 hours, and data-navigation response time from ~1s to 15-20ms.

## IT Support Specialist

Jan 2021 – Sep 2021

RenéDessés de Venezuela — Caracas, Venezuela

- Served as the sole IT/systems administrator for an 80-person company, covering everything from user support to server and network administration.
- Managed Cisco network infrastructure, a MySQL Server database, and the Profit business management system, alongside Apache/Linux servers.
- Performed equipment maintenance and provided ongoing technical support to the company's 80 users.

## FEATURED PROJECTS

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### DropAudio CCS *Nuxt 3 · Supabase · Vercel*

Complete e-commerce platform in production with admin panel, catalog, multi-currency checkout, and interactive audio recommender.

### RealtimeVoiceAssistant *Rust · Python · Gemini Live*

Real-time conversational voice assistant with transcription, NLP, and streaming voice synthesis, built on an async Rust runtime (Tokio).

### System\_Stabilization\_Interpolation *Python · PyTorch · OpenCV — Degree Thesis*

Video stabilization on mobile devices using deep learning and mathematical interpolation; architecture optimized for real-time processing with numerical precision control to avoid critical rounding errors.

### FlowNet Video Stabilization *Python*

Video stabilization system using deep learning: global motion estimation distilled from optical flow.

### extract\_rif *FastAPI · PaddleOCR*

HTTP microservice that extracts structured data from RIF documents (SENIAT) from PDF or image via dual-engine OCR.

### Socket Communication System *Python*

Client-server communication system over TCP sockets with a custom protocol.

## EDUCATION

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### Bachelor's Degree in Computer Science *Graduation: April 2025*

LUZ-IUTA (Universidad del Zulia — University Institute of Technology), Venezuela · Studies: Oct 2023 – Apr 2025 · Program equivalent to a B.Sc. in Computer Science

### Associate Degree (TSU) in Information Technology *2018 – 2021*

LUZ-IUTA, Venezuela

## LANGUAGES

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Spanish — Native · English — B2 (Professional; advanced technical reading/writing, interviews, and professional communication)

## CERTIFICATIONS

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Vue.js 2 Professional Course (Platzi, Jan 2024) · TypeScript: Advanced Types and Functions (Platzi, Jan 2024) · TypeScript Fundamentals (Platzi, Jan 2024) · Web Optimization (Platzi, Aug 2024) · Python: NumPy and Pandas for Data Analysis (Platzi, Aug 2024) · Introduction to Generative AI — Duke University (Coursera, Feb 2025)